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HW#01 – Threat Chain on a Machine Learning System

The incident: In July 2025, investor Jason Lemkin was testing **Replit Agent**, an AI coding agent, on an application with a live production database. After the agent had already made unwanted changes, Lemkin told the AI not to place any change without permission. Even after telling it that, when some db queries returned empty results. Instead of stopping and asking the user what to do, the agent later said it **"panicked"** and ran db commands without permission. Those commands deleted production data containing records for 1,206 executives and more than 1,196 companies.

Fault: AI agent was allowed to directly modify the same db that contained live production data. The code freeze was only an instruction given to the ML agent. It was not enforced by the system's permissions. Therefore, if the agent misunderstood a situation or selected a dangerous action, nothing technically prevented it from changing or deleting production data. The fault was already present before the incident: an unpredictable ML agent had destructive access to a critical resource.

Error: It began when the db queries returned empty results. The agent interpreted those empty results as a problem that it needed to fix. Although it had been told not to make changes, it decided to act and executed destructive db commands. This produced the actual erroneous system state:

the production db contained the required records → those records had been deleted.

So the error was not simply that "the AI made a bad decision." The important error was that the agent's bad decision corrupted the internal state of the production db.

Failure: The failure occurred when that corrupted db could no longer provide the service expected from it. If the application requested one of the deleted records, the correct data was no longer available. The internal db error had therefore escaped into an externally visible service failure.

AI has unsafe production access (Fault) → Agent deletes valid production records (Error) → Application can no longer return the correct data (Failure)



References:

Business Insider, *Replit's CEO apologizes after its AI agent wiped a company's code base in a test run and lied about it*, 22 July 2025.